

Defervescent dengue patients might be a potential source of infection for vector mosquitoes.

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Dengue is a re-emerging public health problem and mosquito-borne infectious disease that is transmitted mainly by *Aedes aegypti* and *Ae. albopictus*. Early diagnosis, isolation, and treatment of patients are critical steps for dengue epidemic control, especially to prevent secondary transmission of dengue virus (DENV). However, little is known about defervescent dengue patients as a source of infection. This case study describes 1268 dengue patients hospitalized at Guangzhou Eighth People's Hospital from June 2013 to December 2014. The viral loads of each individual were measured using real-time reverse transcription-polymerase chain reaction. *Ae. aegypti* and *Ae. albopictus* were exposed to blood meal with gradated dengue viral loads to characterize the relationship between viremia in dengue patients and the vector competence of vector mosquitoes. The viral numbers in the blood were measured, ranging from 108 to 103 copies/ml from day 1 to day 12 after fever onset. Vector competence analysis of *Ae. aegypti* and *Ae. albopictus* indicated that viremia > 104 copies/ml can still infect vector mosquitoes, which implied that the defervescent dengue patients might be a source of infection. The results of this study indicate that some defervescent dengue patients still have sufficient viral load to infect vector mosquitoes. Therefore, the protection against mosquito biting for these people should be extended to prevent secondary transmission events.

Vector competence of the *Aedes aegypti* population from Santiago Island, Cape Verde, to different serotypes of dengue virus.

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Dengue is an arboviral disease caused by dengue virus (DENV), whose main vectors are the mosquitoes *Aedes aegypti* and *Aedes albopictus*. *A. aegypti* is the only DENV vector in Cape Verde, an African country that suffered its first outbreak of dengue in 2009. However, little is known about the variation in the level of vector competence of this mosquito population to the different DENV serotypes. This study aimed to evaluate the vector competence of *A. aegypti* from the island of Santiago, Cape Verde, to four DENV serotypes and to detect DENV vertical transmission. Mosquitoes were fed on blood containing DENV serotypes and were dissected at 7, 14 and 21 days post-infection (dpi) to detect the virus in the midgut, head and salivary glands (SG) using RT-PCR. Additionally, the number of copies of viral RNA present in the SG was determined by qRT-PCR. Furthermore, eggs were collected in the field and adult mosquitoes obtained were analyzed by RT-PCR and the platelia dengue NS1 antigen kit to detect transovarial transmission. High rates of SG infection were observed for DENV-2 and DENV-3 whereas for DENV-1, viral RNA was only detected in the midgut and head. DENV-4 did not spread to the head or SG, maintaining the infection only in the midgut. The number of viral RNA copies in the SG did not vary significantly between DENV-2 and DENV-3 or among the different periods of incubation and the various titers of DENV tested. With respect to DENV surveillance in mosquitoes obtained from the eggs collected in the field, no samples were positive. Although no DENV positive samples were collected from the field in 2014, it is important to highlight that the *A. aegypti* population from Santiago Islands exhibited different degrees of susceptibility to DENV serotypes. This population showed a high vector competence for DENV-2 and DENV-3 strains and a low susceptibility to DENV-1 and DENV-4. Viral RNA copies in the SG remained constant for at least 21 dpi, which may enhance the vector capacity of *A. aegypti* and suggests the presence of a mechanism modulating virus replication in the SG.

The *Aedes aegypti* toll pathway controls dengue virus infection.

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Aedes aegypti, the mosquito vector of dengue viruses, utilizes its innate immune system to ward off a variety of pathogens, some of which can cause disease in humans. To date, the features of insects' innate immune defenses against viruses have mainly been studied in the fruit fly *Drosophila melanogaster*, which appears to utilize different immune pathways against different types of viruses, in addition to an RNA interference-based defense system. We have used the recently released whole-genome sequence of the *Ae. aegypti* mosquito, in combination with high-throughput gene expression and RNA interference (RNAi)-based reverse genetic analyses, to characterize its response to dengue virus infection in different body compartments. We have further addressed the impact of the mosquito's endogenous microbial flora on virus infection. Our findings indicate a significant role for the Toll pathway in regulating resistance to dengue virus, as indicated by an infection-responsive regulation and functional assessment of several Toll pathway-associated genes. We have also shown that the mosquito's natural microbiota play a role in modulating the dengue virus infection, possibly through basal-level stimulation of the Toll immune pathway.

Dengue - Clinical and public health ramifications.

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Dengue virus infection is spread by the mosquito vector *Aedes aegypti* and causes significant morbidity and mortality worldwide. In Australia, it is an important cause of fever in the returned traveller and recent outbreaks have occurred in northern Queensland. A comprehensive understanding of the clinical and public health ramifications of dengue infection is essential for general practitioners. The aim of this article is to review the pathophysiology, clinical manifestations, complications, laboratory investigations and public health consequences of dengue infection. Dengue should be considered as a differential diagnosis of fever in a returned traveller, including in patients who have travelled to northern Queensland within 3 months of an outbreak. Clinical manifestations vary from asymptomatic infection to serious disease. Typical symptoms last 7 days and may include: fever, headache, myalgia, fatigue, abnormal taste sensation, arthralgia, maculopapular rash and anorexia. Around 1% of patients will get the more severe form of the illness, dengue haemorrhagic fever. Recommended diagnostic tests depend on the time since the onset of symptoms. Management involves symptomatic treatment and monitoring for complications. Dengue haemorrhagic fever requires hospitalisation. Prompt notification to public health authorities and advice to patients about prevention of spread are a key role of the GP.

Aedes aegypti Immune Response and Its Potential Impact on Dengue Virus Transmission.

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Dengue virus (DENV) transmission to human populations requires infection of vector mosquitoes as an essential component of the transmission process. DENV transmission leads to infections that range from

asymptomatic to life-threatening pathologies, such as dengue hemorrhagic fever and dengue shock syndrome. *Aedes aegypti* is the principal vector of DENV, and its vector competence consists of the intrinsic factors, genes, molecules, and pathways that allow infection, replication, and dissemination of this virus throughout the cells of mosquito tissues. In the search for mosquito molecular targets to block DENV transmission, the effect of DENV infection on mosquitoes has been an important focus of research. In this study, we review the findings of research on the effect of DENV infection on mosquito tissue cells and the immunity pathways and molecules that are involved in this infection. We emphasize the relevance of recent findings to understand the relationship between *Ae. aegypti* immune response, vector competence, and DENV transmission to human hosts.

Intrinsic features of *Aedes aegypti* genes affect transcriptional responsiveness of mosquito genes to dengue virus infection.

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Dengue virus infection causes significant morbidity and mortality in humans world-wide. The *Aedes aegypti* mosquito is the major vector that spreads dengue virus to humans. Interaction between dengue viruses and *A. aegypti* is a multi-factorial phenomena that is determined by both virus and mosquito genotypes. Although, studies have suggested significant association of mosquito vectorial capacity with population variation of dengue virus, specifications of the vector factors that may influence vector-virus compatibility are very limited in the literature. Recently, we have shown that a large number of genes are differentially expressed between MOYO-S (susceptible) and MOYO-R (refractory) *A. aegypti* strains upon infection with dengue virus (JAM-1409 genotype). In the current study, we show that specific intrinsic features of *A. aegypti* genes are significantly associated with 'responsiveness' of mosquito genes to dengue infection. Binomial logistic regression analysis further reveals differential marginal effects of these features on gene responsiveness of mosquitoes to the viral infection. Thus, our result shows that intrinsic features of genes significantly affect differential expression of *A. aegypti* genes to dengue infection. The information will benefit further investigations on evolution of genes among natural populations of *A. aegypti* conferring differential susceptibility to dengue virus.

Alanine to glycine substitution in the PyR2 confers sodium channel resistance to Type I pyrethroids.

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Aedes aegypti is a primary urban vector of dengue, yellow fever, Zika and chikungunya worldwide. Pyrethroid insecticides are the most effective insecticides for controlling *Ae. aegypti*. However, pyrethroid resistance has developed due to the long-term overuse of the insecticides, and many knockdown resistance (kdr) mutations have been identified in the resistant populations. A1007G, an alanine to glycine substitution, was found in resistant *Ae. aegypti* from Vietnam and Malaysia, which has always co-existed with F1534C and V1016G. However, the role of A1007G in pyrethroid resistance and the linkage of A1007G and F1534C or V1016G remain unknown. In this study, we examined the effects of mutations on the sodium channel gating properties and pyrethroid sensitivity in *Xenopus* oocytes. We found mutations A1007G, A1007G + F1534C and A1007G + V1016G + F1534C shifted the voltage dependence of activation in the depolarizing direction. Mutations A1007G + F1534C and

A1007G + V1016G + F1534C shifted the voltage dependence of inactivation in the depolarizing direction. Both mutations A1007G and F1534C reduced the channel sensitivity to two Type I pyrethroids, permethrin and bifenthrin, and synergistic effects were observed between mutations A1007G and F1534C. However, none of the mutations, A1007G, F1534C and A1007G + F1534C affected the channel sensitivity to two Type II pyrethroids, deltamethrin and cypermethrin. Furthermore, triple mutations A1007G + V1016G + F1534C significantly reduced the channel sensitivity to both Type I and Type II pyrethroids. We identified A1007G had a distinct effect on sodium channel sensitivity to Type I, but not to Type II pyrethroids, also A1007G exhibited synergistic effects with F1534C to Type I pyrethroids, which will provide a fundamental insight into the distinct molecular interactions between insect sodium channel and Type I or Type II pyrethroids. © 2024 Society of Chemical Industry.

Embryonic dormancy in *Aedes aegypti* and *Aedes albopictus* (Diptera: Culicidae): A survival and dispersal mechanism.

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Aedes aegypti and *Aedes albopictus* are the main vectors of arboviruses such as dengue, Zika virus, and chikungunya. *Ae. aegypti* is a widely spread mosquito in tropical and subtropical regions, whereas *Ae. albopictus* is a culicid of Asian origin that shows exophilic behavior and can be found in subtropical and temperate areas. Climatic factors could influence the distribution of both species, making them use genetic and environmental resources to adapt to the environment, activating survival mechanisms (embryonic dormancy) that increase the developmental period and keep their offspring in the environment. This review aimed to compare the different physiological mechanisms of embryonic dormancy between *Ae. aegypti* and *Ae. albopictus* and their impact on the development and environmental adaptability of these two species. A total of 62 articles were identified from PubMed, Scopus, and Web of Science databases corresponding to the period from 1981 to 2021. In diapause, the results mentioned above are indirectly linked to temperature and directly linked to photoperiod variations. With regard to quiescence, temperature and humidity are directly related to the activation of this mechanism. In conclusion, it is essential to highlight the expansion of arboviruses such as dengue, chikungunya, yellow fever, and Zika virus and their relationship with embryonic dormancy, diapause and quiescence which are extremely important strategies for *Ae. aegypti* and *Ae. albopictus* to keep their offspring in the environment under adverse conditions.

Aedes albopictus and the reemergence of Dengue.

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Dengue is a vector-borne disease that is estimated to affect millions of individuals each year in tropical and subtropical areas, and it is reemerging in areas that have been disease-free for relatively long periods of time. In this issue of the journal, Peng et al. report on a Dengue outbreak in a city in southern China that had been disease-free for more than two decades. The infection, which was due to serotype 1, was introduced by a traveler from South-east Asia and transmitted by *Aedes albopictus*, the Asian tiger mosquito. Compared to *Aedes aegypti*, which is the most important vector of Dengue, *Ae. albopictus* is a less competent vector of arboviruses, and the epidemics it causes are milder. However, *Ae. albopictus* is becoming an increasingly important vector because of its rapidly changing global distribution. In

particular, the worldwide trade in second hand tires, which often contain water and are an ideal place for eggs and larvae, has been a key factor in the large-scale conquest of *Ae albopictus*, which easily adapts to new environments, even in a temperate climate. This expansion is creating new opportunities for viruses to circulate in new areas, becoming a common cause of epidemics in *Ae aegypti*-free countries, from Hawaii to Mauritius. The outbreak in China, like similar events, was mild and short-lived. Because epidemics due to *Ae albopictus* are milder, the replacement of *Ae aegypti* with the tiger mosquito could even result in public-health benefits. However, there is no solid evidence of this, and the milder course of the outbreak could be in part explained by the relatively short duration of the hot season in some affected areas. Since it is almost impossible to prevent *Ae albopictus* from being introduced in a country, mosquito-control measures at local level remain the most effective means of controlling arbovirus outbreaks.

A case of dengue from Pune, India.

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Dengue is an acute infectious disease, caused by a single stranded ribonucleic acid (RNA) virus of the family of Flaviviridae, genus Flavivirus, transmitted by *Aedes* mosquitos, the most important vectors being *Aedes aegypti* and *Aedes albopictus*. There are four serotypes, DEN-1, DEN-2, DEN-3, DEN-4 that are nearly human specific. The geographical distribution of dengue is pantropical, except for Madagascar and some African regions. It is endemo-epidemic in tropical and subtropical countries included between 25 degrees north latitude and 25 degrees south latitude, particularly in southeast Asia and epidemic in the Caribbean, West-Africa, tropical American and Pacific Islands. Further spread will depend on the invasion of *Aedes aegypti* into new areas in South America. Maintenance of dengue is supported by an increasing demography, uncontrolled urbanization and climatic conditions favorable to the vectors. Spread of dengue is primarily caused by modern transportation, especially aircraft. No vaccine and no specific treatment exist so that mosquito control is the only way to reduce the incidence of dengue around the world. Dengue is, obviously, an important risk for travelers going to endemic areas.
